MILITARY SPECIFICATION

HYPODERMIC INJECTION TUBE, MORPHINE

This specification has been approved by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers morphine injection for medicinal use in specially designed containers which include collapsible tube with hypodermic needle attached.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

GG-N-196 — Needles; Hypodermic; for Luer Syringes.
PPP-B-566 — Boxes, Folding Paperboard.
PPP-B-585 — Boxes; Wood, Wirebound.
PPP-B-601 — Boxes, Wood, Cleated-Plywood.
PPP-B-621 — Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 — Boxes, Fiber.
PPP-B-676 — Boxes, Set-Up, Paperboard.

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MIL-B-10377 — Box, Wood, Cleated, Veneer, Paper, Overlaid.
MIL-L-10547 — Liners, Case, Waterproof.

STANDARDS

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 — Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

AMERICAN PHARMACEUTICAL ASSOCIATION

National Formulary.

(Application for copies should be addressed to the American Pharmaceutical Association, Washington 7, D. C.)

U. S. PHARMACOPOEIAL CONVENTION, INC.

Pharmacopeia of the United States.

(Application for copies should be addressed to the Mack Publishing Company, Easton, Pa.)

3. REQUIREMENTS

3.1 Material. Material shall be as specified hereinafter.

FSC 6505

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3.1.1 *Solution.* The solution shall contain morphine in water for injection (U.S.P.) and shall be of such concentration that 1.5 cubic centimeters (cc.) shall contain an amount of morphine equivalent to 0.016 gram (Gm.) (¼ grain) of morphine tartrate. The solution shall also contain 0.4 percent phenol, with a tolerance of ±10 percent of specified amount. The formulation shall be as follows:

Morphine ($C_{17}H_{19}O_3N.H_2O$), U.S.P. IX	0.8457 Gm.
Tartaric Acid, N.F., if needed to adjust pH to..	pH 4.55-4.75 in final solution
Phenol, U.S.P.	0.4 Gm.
Water for Injection, U.S.P. .....qs.....	100 cc.

3.1.1.1 *Acidity.* The solution as prepared for filling shall have a pH of 4.55 to 4.75 when determined by use of apparatus employing a glass electrode.

3.2 **Container.** Container shall consist of a collapsible tin tube with a sterile hypodermic needle securely mounted on the neck. A sterile wire stylet for puncturing the inner seal of the tube shall be fitted in place in the cannula of the needle. A transparent plastic hood shall cover the needle and stylet, and shall screw onto the neck of the tube to maintain sterility of the needle and stylet. The overall length of the assembled container shall not exceed 2⅝ inches.

3.2.1 *Tube.* Tube shall be made of block tin, assaying not less than 99.5 percent Sn. The tube shall be ⅜ inch ±1/16 inch in diameter and shall have an overall length of 1½ inches ±⅛ inch to the base of the needle. Neck of the tube shall be 0.180 to 0.184 inch in diameter and 0.208 to 0.216 inch in length. Wall thickness of the tube shall be 0.004 to 0.005 inch. The neck of the tube shall contain an inner seal 0.005 inch to 0.011 inch thick, integral with the neck of the tube and shall be externally threaded to accommodate the cover as specified in 3.2.4. After filling, the lower end of the tube shall be hermetically sealed by welding. The tube shall comply with the requirements of the General Notices section pertaining to containers under Packaging, Storage, and Preservation of the latest revision of the U.S.P.

3.2.2 *Needle.* Needle shall be seamless drawn tubing. Needle and needle hub shall be unplated corrosion-resistant steel, containing from 16 to 20 percent chromium and from 7 to 10 percent nickel (stainless steel). As an alternate material, needle hub only may be made of nickel plated brass. In either case, the hub shall be set in above the tin inner seal of the tube so that it is not in contact with the contents of the tube until after the seal is pierced by the stylet. The outside diameter of the needle shall be 0.035 inch ± 0.001 inch (equivalent to No. 20 gage as specified in Specification GG-N-196). The minimum inside diameter (lumen), measured at point, shall be 0.022 inch. The needle shall be ⅝ inch ±1/16 inch in length above the hub. Hub shall be 0.113 to 0.123 inch outside diameter and 0.036 to 0.040 inch thick and be spun into the neck of the tube. Needle shall have a uniform bore, a sharp bevel point, and shall show a smooth finished outer surface free from pits and tool marks; an inner surface free from scale, deep pits, deep furrows, or die marks; and a lumen and hub free of chips, polisher's dirt, or other foreign material. Bevel shall be 1/64 inch ±1/64 inch in length. The bevel may be ground concave or flat and may have a center ridge at the distal end to strengthen the tip. The bevel shall be so ground at the outer end as to enhance the keenness of the tip and cutting edges. The outer edges of the ground surface shall be keen and both outer and inner edges of the bevel shall be free from burrs and slivers. The tip shall be straight and strong and of maximum keenness.

3.2.2.1 *Additional requirements for needle.* The needle shall meet the tests for corrosion-resistance, stiffness, elasticity, and resistance to breakage as specified in Specification GG-N-196.

3.2.3 *Stylet.* Stylet shall be a wire of unplated corrosion-resistant steel, containing

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from 16 to 20 percent chromium and from 7 to 10 percent nickel (stainless steel). The wire shall have a diameter of 0.02 inch +5 percent and —10 percent tolerance, with one end looped into a circle approximately 5/32 inch in diameter or formed into a triangle approximately 5/32 inch on each side. The stylet shall be of sufficient length and rigidity to permit easy puncture of the inner seal of the tube. A suitable composition washer, approximately 1/8 inch in diameter and 0.020 inch ±10 percent in thickness, shall be used under the looped end of the stylet to protect the fingers from the needle point when puncturing the inner seal.

3.2.4 *Plastic cover.* Cover shall be clear and colorless and of cellulose acetate or equally suitable material. Cover shall be cylindrical in shape, and approximately 1¼ inches in length, and shall be of sufficient wall thickness and strength to prevent distortion, even under warm and humid climatic conditions. The open end shall be internally threaded for affixing to the tube neck. When assembled to the tube, the cover shall be removable from the tube neck by at least one full turn and by not more than two full turns. Cover shall also be removable from the assemblage without, at the same time, pulling the stylet out of the needle.

3.2.5 *Gasket.* Gasket of suitable size and material shall be located on the tube shoulder at base of neck against which to screw the plastic cover.

3.3 **Leakage.** Filled container shall show no leakage of contents when tested as specified in 4.4.1.

3.4 **Security of swaging.** The needle shall be secured to the container so that the cannula shall not be detached from the hub or the hub shall not be detached from the neck of the tube when tested as specified in 4.4.2.

3.5 **Sterility.**

3.5.1 *Morphine bulk solution.* The bulk morphine solution, as specified in 3.1.1, shall be sterile when tested as specified in 4.4.3.1.

3.5.2 *Morphine injection.* The morphine injection from the filled container shall be sterile when tested as specified in 4.4.3.2.

3.5.3 *Needle and stylet.* The needle and stylet shall be sterile when tested as specified in 4.4.3.3.

3.6 **Discharge volume.** Filled container shall deliver not less than 1.5 milliliters (ml.) of morphine injection when tested as specified in 4.4.4. The injection shall be clear, colorless and free from foreign matter, and shall have a pH of 4.55 to 5.20.

3.7 **Stability.**

3.7.1 *Resistance to reduced pressure.* The filled containers shall not crack, break, or leak and the plastic cover shall not warp or expose the sterile parts to possible contamination, when tested as specified in 4.4.5.

3.7.2 *Resistance to cold.* The filled containers shall not crack, break, or leak and the plastic cover shall not warp or expose the sterile parts to possible contamination, when tested as specified in 4.4.6.

3.8 **Assay for morphine.** Morphine injection from the filled containers shall assay not less than 95 percent and not more than 105 percent of morphine equivalent to 16 milligrams (mg.) of morphine tartrate for 1.5 cc., when assayed by the method specified in 4.4.7.

3.9 **Function.** The container shall be so designed and constructed that the stylet shall easily and readily puncture or break the inner seal of the tube without undue use of force.

3.10 **Labeling.**

3.10.1 *Collapsible tube.* The label on the tube shall be printed in red lettering on a white background with the item identifica-

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tion, name of manufacturer, and the following statements: "Sterile", "Warning: May be habit forming", "Read directions for use", and "POISON". Label information shall be lithographed directly onto the metal tube.

3.10.2 *Unit package.*

3.10.2.1 *FSN 6505-129-5517.* The label information shall be printed in red lettering on a white background and shall bear the stock number designated as such, "FSN.....", item identification, lot or control number designated as such, "Lot No. .....", or "Control.....", the name of manufacturer, and the statements: "Warning: May be habit forming", "POISON", and "Sterile". The following directions shall appear on the label or on a separate sheet inserted within the unit package.

"DIRECTIONS FOR USE: Remove transparent hood. Grasp wire loop and push wire in to pierce inner seal, turning if necessary. Pull out and discard wire, thrust needle through skin at least half its length and inject solution by slowly squeezing the tube from the sealed end.

CAUTION: The full effects of morphine may not be felt for 20 to 30 minutes after injection. A second injection of morphine for continued severe pain should not be given sooner than 2 hours after the first one. Never give a second dose of morphine to a person who is breathing twelve or less times a minute. Never give morphine to an unconscious person.

ANTIDOTE: Keep patient awake, do not exhaust him, stimulate respiration, give strong coffee, wash out stomach."

3.10.2.2 *FSN 6505-129-5518.* The label shall be printed in red lettering on a white background and shall bear, in addition to the label information specified in 3.10.2.1, the unit of issue designated as such, "Unit: pkg" and the quantity designated as such, "Quantity: 1". The "DIRECTIONS FOR USE" shall include a schematic diagram of the injection tube with suitable annotations indicating the transparent hood (needle cover), wire loop, and inner seal. The required directions for use shall include the annotations in the proper sequence.

3.10.3 *Additional labeling information.* The label of the collapsible tube and unit package shall not bear the caution labeling as required by the Durham-Humphrey law.

3.11 **Certificate of sterility.** Unless otherwise specified, a certificate of sterility, listing all lot or control numbers, shall state that samples from each sterilizer lot have been tested in accordance with U.S.P. method and found to be sterile and certificate shall be forwarded as designated (see 6.1.1 and 6.2).

3.12 **Workmanship.** The material and its containers shall be free from defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Lot.** The method of lot control numbering shall be placed on file with the procuring activity.

4.2 **Sampling.**

4.2.1 *For visual examination.* From each lot or control number sample size(s) shall be as indicated for inspection level L7 in Appendix to Standard MIL-STD-105.

4.2.2 *For tests.* From each lot or control number randomly select a suitable number of tubes to perform the necessary tests.

4.2.3 If additional sample(s) is required to perform visual examination, as specified in 4.2.1, the additional sample(s) shall be randomly selected, in accordance with Appendix to Standard MIL-STD-105.

4.3 **Inspection.** The inspector shall ascertain that the morphine hypodermic injection tubes have been examined and tested in accordance with specification requirements.

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4.4 Tests.

4.4.1 *Leakage test.* Using a suitable device, a pressure of 15 pounds shall be applied to the tube at an angle of 90° when the tube is on a flat surface or suitably contoured surface to accommodate the tube. The device applying the pressure on the tube shall cover not less than 40 percent of the circumference of the tube and not less than 75 percent of the length of the tube. The 15 pounds shall be applied to the tube for a period of not less than 2 seconds nor more than 4 seconds.

4.4.2 *Swaging test.* With the needle pointing downward, secure the tube to a vise or other suitable device. Attach a total weight of 8 pounds to the cannula by means of clamp, pliers, or other suitable tool, so that the weight pulls downward, away from the tube. Allow the weight to remain suspended from the cannula for a period of 1 minute.

4.4.3 *Sterility.*

4.4.3.1 *Morphine solution.* Morphine solution from the bulk container shall be tested for sterility in the manner specified in the U.S.P. for Sterility Tests for liquids.

4.4.3.2 *Morphine injection.* Morphine injection from the filled containers shall be tested for sterility in the manner specified in the U.S.P. for Sterility Tests for liquids.

4.4.3.3 *Needle and stylet.* Under aseptic conditions, remove the stylets and snip the needles from the filled containers. Immerse 10 stylets and 10 needles in test tubes containing sufficient Fluid Thioglycollate Medium, U.S.P., to cover the needles. Incubate the medium at 30° to 32° centigrade (C.) for 7 days, and examine for viable micro-organisms. No evidence of microbial growth shall be observed.

4.4.4 *Discharge volume.* Discharge volume shall be determined by ejecting contents of 10 hypodermic injection tubes into a graduated cylinder, calibrated to at least 0.1 ml. The total volume shall be not less than 15.00 ml. and no 1 container shall deliver less than 1.35 ml. or more than 1.65 ml. The volume shall be determined between 25° C. and 28° C. The reading shall be made at the bottom of the meniscus (as calibrated) and recorded to the nearest 0.01 ml. The injection shall comply with the requirements of 3.6.

4.4.5 *Resistance to reduced pressure.* Place the filled containers in a vacuum desiccator and maintain at a vacuum of 26 inches of mercury for 10 minutes. Visually examine for signs of damage to determine compliance with 3.7.1.

4.4.6 *Resistance to cold.* Place the filled containers in a suitable refrigerator and refrigerate at minus 30° C. ±2° C. for 6 hours. On return to 25° C. ±5° C., visually examine for signs of damage, particularly the end of the collapsible tube which is closed and sealed by welding. Visually examine for signs of damage to determine compliance with 3.7.2.

4.4.7 *Assay for morphine.* Assay for morphine injection shall be in accordance with the method specified in the latest revision of the U.S.P. for morphine injection.

4.5 Supplier responsibilities for inspection. The supplier shall perform examinations and tests set forth in this specification to substantiate conformance of supplies to specification requirements. The examinations and tests shall be at the expense of the supplier. Suppliers shall have adequate test facilities or shall arrange for test facilities acceptable to the Government to be used for executing specific prescribed tests. Records of examinations and tests performed by the supplier shall be maintained by the supplier and made available to the Government, upon the Government's request, at any time, or from time to time, for a period of 2 years after the delivery of the supplies to which these records relate. The supplier shall be responsible for compliance with all requirements of this specification, and all other technical requirements of the contract.

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5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 *Unit of issue.*

5.1.1.1 *Stock No. 6505-129-5517.* One (1) tube as specified constitutes one unit.

5.1.1.2 *Stock No. 6505-129-5518.* Five (5) tubes as specified constitute one unit.

5.1.2 *Packaging quantities.* The number of units of issue indicated in table I shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in procurement document.

TABLE I. *Packaging Quantities.*

Federal stock number	Unit package	Intermediate package	Shipping container
6505-129-5517	1 unit	50 units	2,500 units
6505-129-5518	1 unit	10 units	500 units

5.1.2.1 *Packing variation permitted.* If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

5.1.3 *Level A.*

5.1.3.1 *Unit package.*

5.1.3.1.1 *Stock No. 6505-129-5517.* Each unit shall be packaged in a paper tube made of spiral wound chipboard, having the following maximum dimensions:

Outside diameter . . 13.3 millimeters (mm.) (0.52 inch)

Wall thickness . . . . 1.0 mm. (0.04 inch) (approx.)

Overall length . . . . 48.0 mm. (1.89 inches)

Each paper tube shall be impregnated with a compound of mineral and vegetable wax at a temperature sufficient to insure complete saturation of the fibers, leaving a minimum of residual wax on the surface. Each paper tube shall be enclosed in a folding carton impregnated with a compound of mineral and vegetable wax at a temperature sufficient to insure complete saturation of the fibers, leaving a minimum of residual wax on the surface. All printing shall be legible through the wax and the maximum overall dimensions of the carton shall be 72 mm. (2.83 inches) long by 15.5 mm. (0.71 inch) by 14.5 mm. (0.57 inch).

5.1.3.1.2 *Stock No. 6505-129-5518.* Each unit shall be packaged in a carton of appropriate size constructed in accordance with Specification PPP-B-676. Carton design shall include suitable interior fittings so that each tube is contained in an individual compartment. Carton shall be adequately secured to prevent accidental opening.

5.1.3.2 *Intermediate package.* Intermediate package shall be a carton of appropriate size constructed in accordance with Specification PPP-B-566 or PPP-B-676. Carton shall be adequately secured to prevent accidental opening.

5.1.4 *Level C.* Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose within the continental limits of the United States of America.

5.2 Packing.

5.2.1 *Level B.*

5.2.1.1 *Exterior container.* Exterior container shall be a double-faced or double-wall corrugated fiberboard carton of suitable size and design, having a minimum bursting strength test of 275 pounds in accordance with Specification PPP-B-636, type I, class 1.

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Double-faced corrugated fiberboard shall be of A or C flutes, double-wall fiberboard of any combination of A, B, or C flutes, except B-B. Box shall be provided with top and bottom pads fabricated from the same strength material as body of box. Box shall be secured.

5.2.2 *Level A.*

5.2.2.1 *Exterior container.* Subject item, packed to the degree of protection required for level B packing, shall be further protected by being overpacked in an exterior container designed for type 2 load and constructed in accordance with Specification PPP-B-585, style 3, for class 3 use; PPP-B-601, table II, using type III, class 1 plywood; PPP-B-621, table III; or MIL-B-10377, table II.

5.2.2.2 *Case liner.* Each level A box shall be lined with a waterproof case liner conforming to Specification MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto.

5.2.2.3 *Strapping.* Strapping, when required, shall be in accordance with applicable box specifications.

NOTE: Strapping will not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.3. *Level C.* The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 *Unit packages.* Each tube and unit carton shall be marked in accordance with applicable requirements of 3.10.

5.3.2 *Intermediate carton.* Intermediate carton shall be marked in accordance with section 5 of Standard MIL-STD-129.

5.3.3 *Shipping container.* Shipping container shall be marked in accordance with Standard MIL-STD-129, except that the word "DRUGS" shall appear in lieu of item identification. Service color and service symbol markings are required only on direct shipments from the contractor's plant to overseas activities.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.

(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.1.1 *Certificate of sterility.* Certificate of sterility shall be furnished unless the procuring activity expresses a desire to omit the certificate (see 3.11).

6.2 For Military Medical Supply Agency procurement.

6.2.1 *Source inspection.* When inspection is made at source, certificate of sterility, required in 3.11, shall be submitted to cognizant inspector for transmittal to Head, Quality Control Division, Technical Department, Military Medical Supply Agency, 84 Sands Street, Brooklyn 1, New York.

6.2.2 *Destination inspection.* When inspection is made at destination, certificate of sterility, required in 3.11, shall be submitted by contractor directly to Head, Quality Control Division, Technical Department, Military Medical Supply Agency, 84 Sands Street, Brooklyn 1, New York, and a copy shall be forwarded with shipment to the consignee.

6.3 This specification covers the following items appearing in the Armed Services Medical Stock List:

6505-129-5517 MORPHINE INJECTION, USP, 16 mg. (¼ gr.).

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6505-129-5518 MORPHINE INJEC. TION, USP, 16 mg. (¼ gr.), 5s.

6.4 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of this specification, or which are commercially available, but is intended to cover the types, etc., which are normally procured to meet Military requirements.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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Navy—Bureau of Medicine and Surgery
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